

18) Suppose P is invertible and $A = PBP^{-1}$. Solve for B in terms of A .

Solution: By the defn. of invertible matrix, there exists a unique matrix P^{-1} such that $P^{-1}P = PP^{-1} = I$.

$$\begin{aligned} A &= PBP^{-1} \Rightarrow AP = (PBP^{-1})P \Rightarrow AP = (PB)(P^{-1}P) \text{ (Theorem 2(a))} \\ &\Rightarrow AP = (PB)I \text{ (By the defn. of inverse)} \\ &\Rightarrow AP = PB \text{ (Theorem 2(e))} \Rightarrow P^{-1}(AP) = P^{-1}(PB) \\ &\Rightarrow (P^{-1}A)P = (P^{-1}P)B \text{ (Theorem 2(a))} \Rightarrow P^{-1}AP = IB \text{ (By the defn. of inverse)} \\ &\Rightarrow P^{-1}AP = B \text{ (Theorem 2(e))} \end{aligned}$$

$$\boxed{B = P^{-1}AP} \quad \square$$

19) If A, B and C are $n \times n$ invertible matrices, does the equation $C^{-1}(A+X)B^{-1} = I_n$ have a solution, X ? If so, find it.

Solution:

$$\begin{aligned} C^{-1}(A+X)B^{-1} &= I_n \Rightarrow (C^{-1}(A+X)B^{-1})B = I_n B \\ &\Rightarrow (C^{-1}(A+X))(B^{-1}B) = I_n B \text{ (Theorem 2(a))} \\ &\Rightarrow (C^{-1}(A+X))I = I_n B \text{ (By the defn. of inverse)} \\ &\Rightarrow (C^{-1}(A+X)) = B \text{ (Theorem 2(e))} \\ &\Rightarrow C(C^{-1}(A+X)) = CB \Rightarrow (CC^{-1})(A+X) = CB \text{ (Theorem 2(a))} \\ &\Rightarrow I(A+X) = CB \text{ (By the defn. of inverse)} \\ &\Rightarrow (A+X) = CB \text{ (Theorem 2(e))} \\ &\Rightarrow -A + (A+X) = -A + CB \\ &\Rightarrow (-A+A) + X = -A + CB \text{ (Theorem 2(a))} \end{aligned}$$

$$\begin{aligned} -A + A &= [-a_1 \ -a_2 \ \dots \ -a_n] + [a_1 \ a_2 \ \dots \ a_n] \text{ Matrix} \\ &= [(-a_1+a_1) \ (-a_2+a_2) \ \dots \ (-a_n+a_n)] \text{ (vector addition)} \\ &= [0 \ 0 \ \dots \ 0] \text{ (Algebraic property iv)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} 0 + X &= -A + CB \Rightarrow X + 0 = CB + (-A) \text{ (Theorem 1(a))} \\ &\Rightarrow X = CB - A \text{ (Theorem 1(c) and defn. of subtraction of matrix)} \end{aligned}$$

Verification of whether $X = CB - A$ is a solution

$$\begin{aligned} C^{-1}(A + (CB - A))B^{-1} &= C^{-1}(A + (-A + CB))B^{-1} \text{ (Theorem 2(a))} \\ &= C^{-1}(A + (-A) + CB)B^{-1} \text{ (Theorem 1(b))} \end{aligned}$$